

The empirical recurrence for OEIS A251200, recovered from its tail

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8 September 2026

Abstract

OEIS A251200 records “Empirical recurrence of order 56” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 256$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 6$, so the recurrence is proved for every $n \geq 62$. Its last coefficient is $c_{56} = -9 \neq 0$, so no further tail satisfies anything shorter; any order-56 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A251200 is “Number of $(n+1) \times (7+1)$ 0..1 arrays with no 2×2 subblock having the minimum of its diagonal elements less than the absolute difference of its antidiagonal elements.”. It has offset 1 and begins

2776, 56519, 1470646, 40413851, 1123772109, 31288421074, 871573835362, 24277088319379, ...

The entry states:

Empirical recurrence of order 56 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 13:10:25, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 256$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 256$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 56: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 6$ is the first whose minimal order is exactly the stated 56.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 512$ exact terms from $n = 6$ on, it returns order 56 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{56} c_i a(n-i),$$

c_1	2	c_{15}	−2324120611138	c_{29}	−862025955224164	c_{43}	−175813703395
c_2	695	c_{16}	−5375866057809	c_{30}	−1510591298539864	c_{44}	292215948838
c_3	4241	c_{17}	21204618265475	c_{31}	873637447978046	c_{45}	41147979917
c_4	−87886	c_{18}	2934095715020	c_{32}	1127467686788159	c_{46}	−17353395943
c_5	−533589	c_{19}	−93084467297703	c_{33}	−595972847257446	c_{47}	−3866963864
c_6	6000022	c_{20}	86161567341982	c_{34}	−638339049526459	c_{48}	502693627
c_7	25592803	c_{21}	194014629912995	c_{35}	270693519957689	c_{49}	183472245
c_8	−260632999	c_{22}	−386123296944261	c_{36}	272110302046645	c_{50}	−974568
c_9	−524968042	c_{23}	−138033565240799	c_{37}	−78992978326438	c_{51}	−3970431
c_{10}	7080807855	c_{24}	833640033403520	c_{38}	−85553682561659	c_{52}	−200733
c_{11}	1353239703	c_{25}	−156993855219557	c_{39}	13277034878833	c_{53}	34772
c_{12}	−115395345512	c_{26}	−1267212694242517	c_{40}	19182483916240	c_{54}	2690
c_{13}	133810776827	c_{27}	556518210284848	c_{41}	−708337406181	c_{55}	−97
c_{14}	1044913283247	c_{28}	1552221745145490	c_{42}	−2940139924226	c_{56}	−9

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 62$, and it is the recurrence OEIS A251200 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 6$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{56} - c_1 x^{55} - \dots - c_{56}$. Its constant term is $-c_{56} = 9$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 56 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 56, so $\deg q = 56$, and it is monic. It holds from some index t on. From $\max(t, 6)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 56, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 14 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 56, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -9 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-56 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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